

- 1) A satellite's onboard computer keeps track of how the satellite is oriented relative to the stars (i.e., how the satellite's body-fixed coordinate system $(\mathcal{B} : \{O, \hat{b}_1, \hat{b}_2, \hat{b}_3\})$ is oriented with respect to the inertial or Newtonian system $(\mathcal{N} : \{O, \hat{n}_1, \hat{n}_2, \hat{n}_3\})$. The satellite needs to transmit this information frequently to Earth. In order to reduce the amount of data transmitted, the satellite is transmitting just part of the direction cosine matrix (DCM), $C_{\mathcal{B}\mathcal{N}}$ that maps a vector in the $\mathcal{N}$ -frame to the vector in the $\mathcal{B}$ -frame at a specific time instant, t . You are part of the ground station team and your task is to reconstruct the entire matrix.

- a) Find the values of the missing elements a , b , and c . You may assume $C_{\mathcal{B}\mathcal{N}}$ is a proper rotation, i.e., $\det(C_{\mathcal{B}\mathcal{N}}) = +1$.

$$C_{\mathcal{B}\mathcal{N}} = \begin{bmatrix} 0.5826 & 0.6943 & 0.4226 \\ a & 0.5371 & 0.2346 \\ b & c & 0.8754 \end{bmatrix}$$

- b) Determine the angle between the satellite's 1-axis $\hat{b}_1$ and the Newtonian 2-axis $\hat{n}_2$ at the same time instant t .
- 2) A small satellite is equipped with two attitude sensors: a Sun sensor that measures the direction to the Sun in the body frame, and a three-axis magnetometer that measures the Earth's magnetic field direction in the body frame. At a given instant, the ground team's ephemeris and geomagnetic model give the corresponding directions in the ECI frame $\mathcal{N}$:

$$\mathcal{N}\hat{s} = \begin{bmatrix} 0.6 \\ 0.8 \\ 0 \end{bmatrix} \quad (\text{Sun direction}), \quad \mathcal{N}\hat{m} = \begin{bmatrix} 0 \\ 0.6 \\ 0.8 \end{bmatrix} \quad (\text{magnetic field direction}).$$

At the same instant the onboard sensors read

$$\mathcal{B}\hat{s} = \begin{bmatrix} 0.8573 \\ 0.1414 \\ 0.4950 \end{bmatrix}, \quad \mathcal{B}\hat{m} = \begin{bmatrix} -0.0326 \\ 0.4243 \\ 0.9050 \end{bmatrix}.$$

The *TRIAD algorithm* constructs an orthonormal triad from each pair of vectors and then reads off the DCM directly. Given two non-parallel unit vectors $\hat{s}$ and $\hat{m}$ (expressed in some frame), define the triad

$$\hat{t}_1 = \hat{s}, \quad \hat{t}_2 = \frac{\hat{s} \times \hat{m}}{\|\hat{s} \times \hat{m}\|}, \quad \hat{t}_3 = \hat{t}_1 \times \hat{t}_2.$$

- a) Compute the triad $\{\hat{t}_1^N, \hat{t}_2^N, \hat{t}_3^N\}$ from the inertial-frame measurements, and assemble the matrix $T_N = [\hat{t}_1^N \ \hat{t}_2^N \ \hat{t}_3^N]$ (columns are the triad vectors in $\mathcal{N}$ -frame components).
- b) Repeat with the body-frame measurements to obtain $T_B = [\hat{t}_1^B \ \hat{t}_2^B \ \hat{t}_3^B]$.
- c) Show that $C_{\mathcal{B}\mathcal{N}} = T_B T_N^\top$, and use this to compute the satellite's attitude $C_{\mathcal{B}\mathcal{N}}$. Verify that your result is orthogonal to within rounding.
- d) As a physical consistency check, compute the angle between $\hat{s}$ and $\hat{m}$ in the inertial frame and in the body frame. Should they agree exactly? Explain briefly.

TRIAD is what every CubeSat with a Sun sensor and magnetometer runs (in some form) for coarse attitude determination.

- 3) Two spacecraft, the Chaser and the Target, are in close proximity during a rendezvous maneuver in low Earth orbit as shown in Figure 1. The Target defines a Local-Vertical Local-Horizontal (LVLH) frame $(\mathcal{T} : \{\hat{T}, \hat{t}_1, \hat{t}_2, \hat{t}_3\})$ with axes:

$$\hat{t}_1 = \frac{\vec{r}_T}{\|\vec{r}_T\|}, \quad \hat{t}_3 = \frac{\vec{r}_T \times \vec{v}_T}{\|\vec{r}_T \times \vec{v}_T\|}, \quad \hat{t}_2 = \hat{t}_3 \times \hat{t}_1.$$

Similarly, the Chaser defines its own LVLH frame ($\mathcal{C} : \{C, \hat{c}_1, \hat{c}_2, \hat{c}_3\}$).

At a given instant, the states in the inertial frame ($\mathcal{N} : \{O, \hat{n}_1, \hat{n}_2, \hat{n}_3\}$) are:

$$\begin{aligned}\mathcal{N}\vec{r}_T &= -6700\hat{n}_1 + 2500\hat{n}_2 + 100\hat{n}_3 \text{ (in Km)}, & \mathcal{N}\vec{v}_T &= -2.80\hat{n}_1 - 7.00\hat{n}_2 + 0.05\hat{n}_3, \text{ (in Km/s)} \\ \mathcal{N}\vec{r}_C &= -6650\hat{n}_1 + 2600\hat{n}_2 + 200\hat{n}_3 \text{ (in Km)}, & \mathcal{N}\vec{v}_C &= -2.85\hat{n}_1 - 6.98\hat{n}_2 + 0.08\hat{n}_3 \text{ (in Km/s)},\end{aligned}$$

- Construct the DCM which maps a vector in the $\mathcal{N}$ -frame to a vector in the $\mathcal{T}$ frame.
- Construct the DCM which maps a vector in the $\mathcal{N}$ -frame to a vector in the $\mathcal{C}$ frame.
- Determine the DCM that transforms from the Chaser's frame to the Target's frame.
- Interpret the result: quantify the total angular misalignment between the Chaser and Target frames (in degrees, and in arcseconds). Which axis is most misaligned, and does the magnitude of the misalignment agree with your physical intuition given the small separation between the two spacecraft?

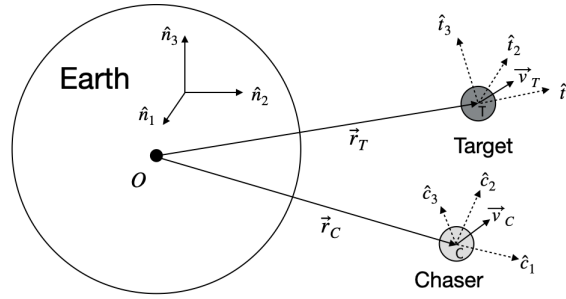


Fig. 1. A Schematic of Rendezvous Maneuver

- Two common reference frames used in the field of astrodynamics are the Earth-Centered Inertial (ECI) and Earth-Centered Earth-Fixed (ECEF) frames (cf. Figure 2). The ECI frame ($\mathcal{N} : \{O, \hat{n}_1, \hat{n}_2, \hat{n}_3\}$) is defined such that the center of the Earth coincides with the origin, the $\hat{n}_3$ axis is along the Earth's axis of rotation, and the $\hat{n}_1$ -axis points towards vernal equinox¹ (in other words, it does not rotate with the Earth). The ECEF frame ($\mathcal{E} : \{O, \hat{e}_1, \hat{e}_2, \hat{e}_3\}$), is defined such that it matches the rotation of the Earth, with the $\hat{e}_1$ -axis points towards the intersection of the equator and the prime meridian², essentially toward the Greenwich. The Earth rotates about $\hat{e}_3$ -axis with angular velocity $\omega^{\mathcal{E}/\mathcal{N}} \approx 7.29 \times 10^{-5}$ rad/s. Assume that at $t = 0$ the 1-axes of the ECI and ECEF frames are aligned.
 - Derive the DCM that maps a vector in the $\mathcal{N}$ -frame to a vector in the $\mathcal{E}$ frame at a time instant, t .
 - Compute the DCM that maps a vector in the $\mathcal{N}$ -frame to a vector in the $\mathcal{E}$ frame at $t = 7$ h 23 min 15 s. Verify that the DCM is orthogonal.
 - An astronomer on Earth wants to point a telescope at a fixed distant star whose direction in the ECI frame is $\mathcal{N}\hat{s} = [0.5, 0.7, 0.51]^T$. At what time $t \in [0, 24$ h] (give all solutions to the nearest second) does the direction from the origin along the ECEF $\hat{e}_1$ -axis first align with the projection of $\hat{s}$ into the equatorial plane?
- Once again, consider the ECI frame ($\mathcal{N} : \{O, \hat{n}_1, \hat{n}_2, \hat{n}_3\}$). As shown in Figure 3, a topocentric frame ($\mathcal{T} : \{T, \hat{t}_1, \hat{t}_2, \hat{t}_3\}$) can be defined for a ground station with specified latitude and longitude (ϕ, λ) such that the basis vectors $\hat{t}_1, \hat{t}_2$, and $\hat{t}_3$, are pointing up at zenith, north, and east, respectively. Define an intermediate frame ($\mathcal{E} : \{O, \hat{e}_1, \hat{e}_2, \hat{e}_3\}$) such that the basis vector $\hat{e}_3 = \hat{n}_3$ and $\hat{e}_1$ points along the equatorial plane from the center of the Earth to the great circle corresponding to the longitude of the ground station. At a given time, t , the angle between $\hat{e}_1$ and $\hat{n}_1$ will be $\theta = \theta_0 + \omega^{\mathcal{E}/\mathcal{N}}t$, with θ_0 being an initial offset and $\omega^{\mathcal{E}/\mathcal{N}}$ the angular velocity of the Earth about the $\hat{n}_3$ axis.

¹Intersection of Earth's equatorial plane and the mean ecliptic on January 1, 2000.

²The prime meridian is the line of 0° longitude which passes through Greenwich, England

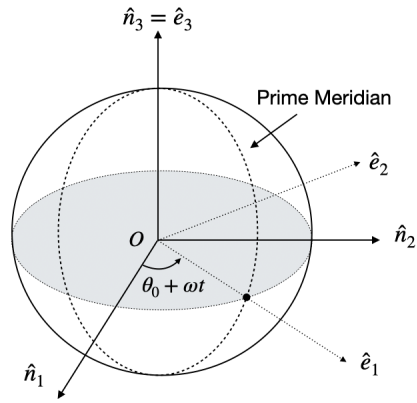


Fig. 2. A Schematic of ECI and ECEF Frames

- a) Assuming that the Earth is a perfect sphere, the transformation between the inertial and topocentric frames can be summarized as ${}^T\vec{r} = C_{TN}{}^N\vec{r} - R_{\oplus}\hat{t}_1$, where $R_{\oplus}$ is the radius of the Earth, ${}^N\vec{r}$ is a position vector of a satellite in the ECI frame and ${}^T\vec{r}$ is a position vector of a satellite in the topocentric frame. Find C_{TN} in terms of ϕ , λ , and θ (**Hint:** the DCM C_{TN} can be decomposed into a rotation about $\hat{n}_3$ followed by a rotation about $\hat{e}_2$).
- b) Let $R_{\oplus} = 6370$ km, $\theta_0 = 0$, and $\omega^{\mathcal{E}/\mathcal{N}} \approx 7.29 \times 10^{-5}$ rad/s. The position of a satellite in the inertial frame at time $t = 500$ s is ${}^N\vec{r} = [1400, 400, 6800]^T$ km, find the position of the satellite in the topocentric frame for the SvalSat ground station at Svalbard $(\phi, \lambda) = (78.23^\circ\text{N}, 15.65^\circ\text{E})$, one of the world's busiest ground stations for polar-orbiting satellites.
- c) Determine if the satellite is above the local horizon of the ground station.
- d) If the satellite is visible, compute its *elevation angle* above the horizon and its *azimuth* measured clockwise from north (i.e., $\text{az} = 0^\circ$ is north, 90° is east). These are the pointing commands SvalSat would send to its dish.

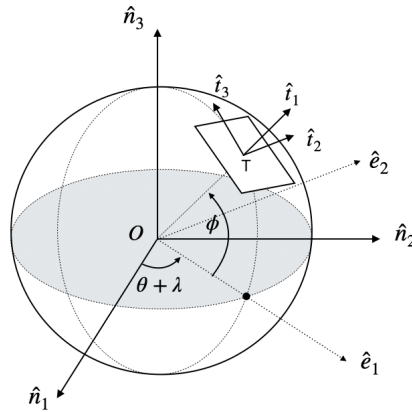
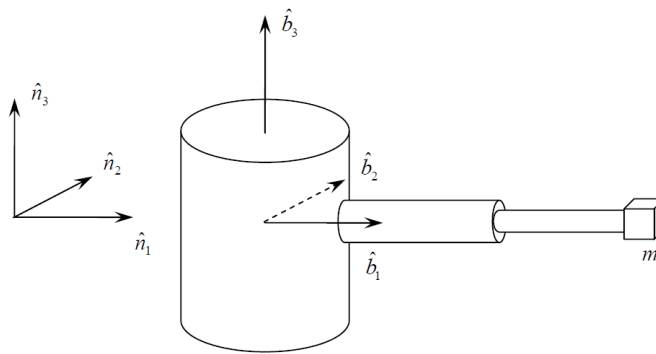


Fig. 3. A Schematic of Topocentric Frame

- 6) During a ground test, a cylindrical-shaped satellite is rotating at constant angular rate ω about the $\hat{b}_3$ -axis. At time $t = 0$, the body fixed frame $\mathcal{B}$ is aligned with the Newtonian frame $\mathcal{N}$ (i.e., $\hat{b}_i = \hat{n}_i$, $i = 1, 2, 3$) as shown in the figure below. A motor is driving a telescoping arm outward along the $\hat{b}_1$ -axis, such that at time t , the mass m at the tip of the arm is located at $[k_1 + k_2 \sin(\Omega t)]\hat{b}_1$, where Ω , k_1 and k_2 are positive constants with $k_1 > k_2$ (so the arm length is always positive).



- a) Find an expression for the inertial velocity of mass m , expressed in terms of the $\hat{b}_i$ vectors, as a function of time.
 - b) Find an expression for the DCM $C_{B\mathcal{N}}$ that maps from the $\mathcal{N}$ -frame to the $\mathcal{B}$ -frame.
 - c) Find an expression for the inertial velocity of mass m , expressed in terms of the $\hat{n}_i$ vectors, as a function of time.
- 7) Consider small sounding rockets that are to be launched from each of the following three locations (look up geographic coordinates as needed):
- Vandenberg Space Force Base, California, USA
 - Baikonur Cosmodrome, Kazakhstan
 - Guiana Space Centre (Centre Spatial Guyanais), Kourou, French Guiana
- a) In each case, calculate the speed of the vehicle just prior to launch in a Newtonian reference frame (i.e., one that does not rotate with respect to the stars) with origin at the Earth's center. Use $R_{\oplus} = 6378$ km and $\omega^{\mathcal{E}/\mathcal{N}} = 7.2921 \times 10^{-5}$ rad/s.
 - b) For a rocket launched due east from each site, the Earth's rotation contributes a “free” velocity boost toward orbit. Rank the three sites by the size of this boost and comment on the pattern.
 - c) Vandenberg is used almost exclusively for polar and retrograde (sun-synchronous, high-inclination) launches, while Kennedy Space Center and Kourou are the go-to sites for prograde equatorial launches. Using your results, explain the mission-design reason for this specialization.